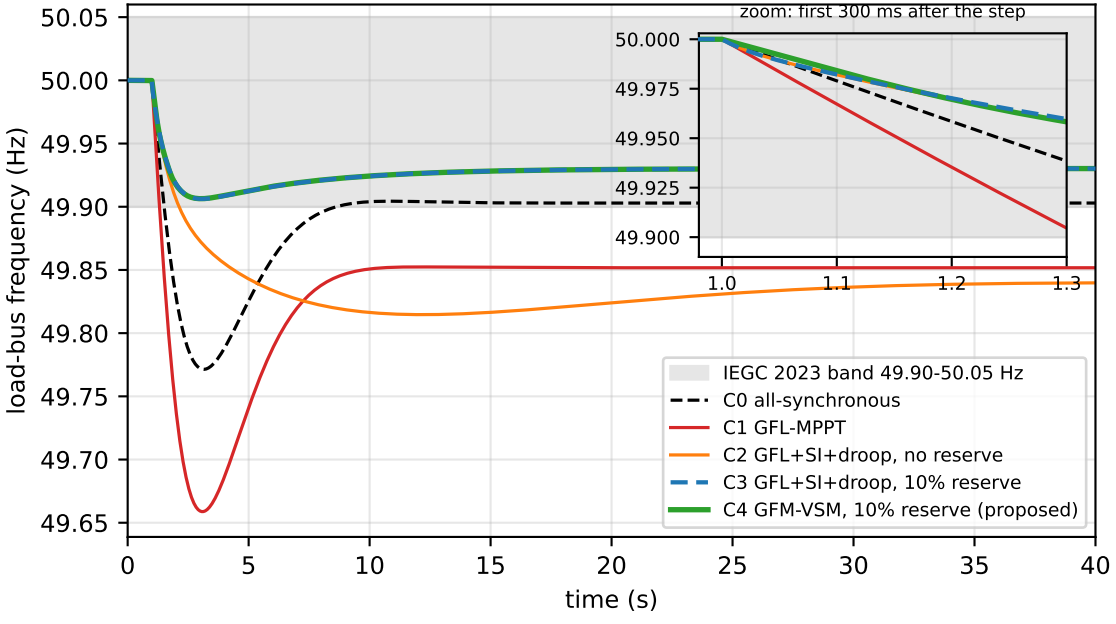
50 MW load step at t = 1 s, wind penetration 40%, 9 m/s
load-bus frequency (Hz)
time (s)
50.05
50.00
49.95
49.90
49.85
49.80
49.75
49.70
49.65
0
5
10
15
20
25
30
35
40
zoom: first 300 ms after the step
50.000
49.975
49.950
49.925
49.900
1.0
1.1
1.2
1.3
IEGC 2023 band 49.90-50.05 Hz
C0 all-synchronous
C1 GFL-MPPT
C2 GFL+SI+droop, no reserve
C3 GFL+SI+droop, 10% reserve
C4 GFM-VSM, 10% reserve (proposed)